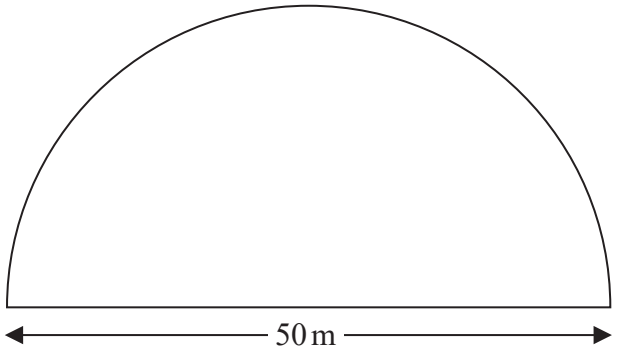
19 A farmer has a field in the shape of a semicircle of diameter 50 m.



The farmer asks Jim to build a fence around the edge of the field.
Jim tells him how much it will cost.

Total cost = £29.86 per metre of fence plus £180 for each day's work

Jim takes three days to build the fence.

Work out the total cost.

Curved edge = ½ × π × 50 = 25π m

1 mark
Half a circumference

Total fence length = 25π + 50 m

1 mark
Include the straight diameter

	m	GBP
×50	1	29.86
×50	50	1493

Straight fence: £1493

Curved fence: £(25π × 29.86)

Labour = 3 × £180 = £540

1 mark
Find costs for different aspects

Total = 1493 + 25π × 29.86 + 540

1 mark
Add fencing and labour costs

= 4378.198915...

Cost = £4378.20

1 mark
Round money at the end

£ 4378.20

(Total for Question 19 is 5 marks)